

## Peer Response (Fahad)

by [Ali Alzahmi](#) - Sunday, 10 August 2025, 12:41 PM

Your post on agent-based systems (ABS) offers a comprehensive and thought-provoking insight into how agent-based systems are a radical move from monolithic to decentralised, adaptive, and intelligent system structures. I especially like how you linked the creation of ABS with the requirements to deal with complex, non-linear organisational ecosystems where more top-down control is less applicable (Dhasarathan et al., 2024). How you incorporated examples like algorithmic trading efficiently, managing traffic, and resilient supply chains makes it a practical reality, and it is easy to envisage how ABS can be applied to different industries. The second powerful aspect mentioned was the inclusion of emergent behavior as an essential property, which would reflect the distinct strength that ABS has over other methods in simulating complex interdependencies and unpredictability (Pulikottil et al., 2021). Your argument is further reinforced by your description of distributed cognition and its connection with decentralised decision making, demonstrating how ABS can utilise collective intelligence to achieve positive results.

Although your post is excellent in displaying the advantages, it would be better to give some space to explaining the difficulties as well. As an example, the very form of emergent behaviors that introduce adaptability would, under some circumstances, introduce unpredictability and necessitate governance and monitoring in turn. Moreover, your future-oriented views would be augmented by adding a holiday message on how ABS interacts with other emerging technologies, including blockchain, advanced analytics, or IoT (Gurcan, 2024). This would aid in demonstrating that ABS is not some isolated innovation but a part of a digital transformation. Your overall analysis is detailed and structured, clearly showing why ABS adoption is a technical upgrade and strategic organisational step (Dhasarathan et al., 2024). It conveys the operational and strategic benefits, albeit in a minimal way that will allow us to delve deeper into integration and governance issues later.

## References

Dhasarathan, C., Shanmugam, M., Kumar, M., Tripathi, D., Khapre, S., & Shankar, A. (2024). A nomadic multi-agent based privacy metrics for e-health care: a deep learning approach. *Multimedia Tools and Applications*, 83(3), 7249-7272. Available at: <https://doi.org/10.1007/s11042-023-15363-4> (Accessed: 09 August 2025).

Gurcan, O. (2024). LLM-Augmented Agent-Based Modelling for Social Simulations: Challenges and Opportunities. *ArXiv*. Available at: <https://arxiv.org/abs/2405.06700> (Accessed: 09 August 2025).

Pulikottil, T., Estrada-Jimenez, L. A., Rehman, H. U., Barata, J., Nikghadam-Hojjati, S., & Zarzycki, L. (2021, September). Multi-agent based manufacturing: current trends and challenges. In *2021 26th IEEE international conference on emerging technologies and factory automation (ETFA)* (pp. 1-7). IEEE. Available at: 10.1109/ETFA45728.2021.9613555 (Accessed: 09 August 2025).

## Peer Response (Rayyan)

by [Ali Alzahmi](#) - Sunday, 17 August 2025, 9:13 PM

Dear Rayyan,

The adaptability and decentralised structure of agent-based systems (ABS) are emphasised in your post. I want to go into more detail about how these systems are used in the healthcare industry and how they fit with the idea that organizations are complex adaptive systems.

ABS are being used more and more in the healthcare industry to enhance patient service delivery, system coordination, and data integration. Because healthcare environments are dynamic and highly distributed, requiring interaction between numerous systems, devices, and stakeholders, agents are able to handle tasks like scheduling appointments, monitoring patients, and coordinating emergency responses on their own. A mobile multi-agent system that facilitated safe patient data exchange and decision-making in telemedicine settings was created by Karami and Hosseini Shahmirzadi (2018).

Additionally, complexity science is the foundation of ABS. Organisations can be thought of as complex adaptive systems in which emergent, system-wide behaviors are produced by local interactions between people or departments. These behaviors can be simulated using agent-based modeling, which aids leaders in testing plans and anticipating unforeseen outcomes. In situations that are changing quickly, this promotes improved decision-making and adaptability (Gómez-Cruz et al., 2017).

These viewpoints reaffirm that ABS is a conceptual framework for managing and modeling complexity in systems and organisations, in addition to being a technical solution. As industries adopt digital transformation and automation powered by AI, their significance only increases.

## References

Gómez-Cruz, N.F., Loaiza Saa, A. and Ortega Hurtado, E., 2017. Understanding organisational dynamics through agent-based modelling. *European Journal of Management and Business Economics*, 26(3), pp.313–331. Available at: <https://www.emerald.com/insight/content/doi/10.1108/EJMBE-10-2017-018/full/html> (Accessed: 17 August 2025).

Karami, A. and Hosseini Shahmirzadi, M.A., 2018. Mobile multi-agent systems in healthcare: a comprehensive review. *Journal of Biomedical Physics and Engineering*, 8(3), pp.315–326. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5971188> (Accessed: 17 August 2025).

## Peer Response (Koulthoum)

by [Ali Alzahmi](#) - Sunday, 10 August 2025, 12:39 PM

In your post, you have presented an easy-to-read and understandable concept of the reasons behind the popularity of adopting agent-based systems (ABS), especially in connection with the distributed computing sphere, the application of AI, and the level of increasing discrimination within organisational settings. I found it interesting that you defined ABS as systems and that they were required to be autonomous systems, responsive, and in turn can initiate their interactions on shared ecosystems (Sicard et al., 2021). The comparison you made between the centralised old-school systems and decentralized ABS is a good point because it helps reveal how the latter may better fit the existing world and real-life circumstances, particularly in logistics, smart grids, and autonomous vehicles. Using the case of individuals serving as the agents of various entities to pursue individual and collectively oriented goals was an excellent idea to bring the concept down to the ground (Berger et al., 2024).

The advantages of ABS, such as being scalable, fault-tolerant, and flexible, are some of the organisational strengths of your post. Another good argument that you make is on the aspect of adaptability in competition and uncertain environments. To add to this, you may speak about possible drawbacks or problems, i.e., how these systems deal with conflicting interests among the agents or whether there is a risk of a coordination failure when scaling up (Cardoso & Ferrando, 2021). Future-focused aspects like how ABS could tie into predictive analytics, digital twins, or state-of-the-art simulation space would have improved your conclusion. Not only is the core idea of ABS in contemporary contexts being conveyed in your post, but also, with the acknowledgment of implementation difficulties as well as advantages, the post might provide a more complete view on their use in the creation of the future of intelligent, decentralised organisational systems (Berger et al., 2024).

## References

Berger, U., Bell, A., Barton, C. M., Chappin, E., Dreßler, G., Filatova, T., Fronville, T., Lee, A., Van Loon, E., Lorscheid, I., Meyer, M., Müller, B., Piou, C., Radchuk, V., Roxburgh, N., Schüler, L., Troost, C., Wijermans, N., Williams, T. G., . . . Grimm, V. (2024). Towards reusable building blocks for agent-based modelling and theory development. *Environmental Modelling & Software*, 175, 106003. Available at: <https://doi.org/10.1016/j.envsoft.2024.106003> (Accessed: 10 August 2025).

Cardoso, R. C., & Ferrando, A. (2021). A Review of Agent-Based Programming for Multi-Agent Systems. *Computers*, 10(2), 16. Available at: <https://doi.org/10.3390/computers10020016> (Accessed: 10 August 2025).

Sicard, V., Andraud, M., & Picault, S. (2021). Organization as a Multi-level Design Pattern for Agent-based Simulation of Complex Systems. In *ICAART (1)* (pp. 232-241). Available at: <https://www.scitepress.org/PublishedPapers/2021/102232/102232.pdf> (Accessed: 10 August 2025).

## Peer Response (Ali Alhammadi)

by [Ali Alzahmi](#) - Sunday, 10 August 2025, 12:37 PM

In your post, you present a very balanced and comprehensive introduction to agent-based systems (ABS), with its obvious beginning presented in contrast to the more commonplace standard and rigid understanding of centralised modes of systematisation. I also liked how you showed the importance of artificial intelligence methods to empower the agents to act autonomously and adjust to their world: machine learning, logical reasoning, and behavior modelling (Axtell & Farmer, 2025). Such interrelation of ABS and AI is paramount, since it reveals that even though their emergent usage is seen as a form of decentralisation, they are also about greater intelligence and adaptability. The way you addressed every single one of the described advantages (flexibility, scalability, fault tolerance, decentralised decision-making) was quite solid. You correlated these improvements to real-life practices related to supply chains, robotics, financial markets, and healthcare, making the argument more practical (Ghaffarzadegan et al., 2024).

Another strength was your explanation of the various models of agents, reactive, deliberative, and the hybrid, and how they are used in simulations, which depicts the flexibility of ABS in the representation of complex systems. To increase the strength of your analysis, you might identify any difficulty regarding the implementation, like the impossibility of coordinating with many agents working independently or the aspects of governance relative to transparency and accountability in the decision process (Sakurada et al., 2022). Also, it can be beneficial to elaborate more on how multi-agent systems with other emerging technologies, such as IoT, cloud computing, or blockchain, might make them more efficient and trustworthy (Ghaffarzadegan et al., 2024). Generally, your answer aptly expresses the idea behind ABS's conceptual framework and practical consequences. It offers a flexible, forward-looking solution to cope with complexity in the contemporary organisational world.

## References

Axtell, R. L., & Farmer, J. D. (2025). Agent-based modeling in economics and finance: Past, present, and future. *Journal of Economic Literature*, 63(1), 197–287. Available at: <https://doi.org/10.1257/jel.20221319> (Accessed: 10 August 2025).

Ghaffarzadegan, N., Majumdar, A., Williams, R., & Hosseinichimeh, N. (2024). Generative agent-based modeling: An introduction and tutorial. *System Dynamics Review*, 40(1), e1761. Available at: <https://doi.org/10.1002/sdr.1761> (Accessed: 10 August 2025).

Sakurada, L., Leitao, P., & La Prieta, F. D. (2022). Agent-Based Asset Administration Shell Approach for Digitizing Industrial Assets. *IFAC-PapersOnLine*, 55(2), 193-198. Available at: <https://doi.org/10.1016/j.ifacol.2022.04.192> (Accessed: 10 August 2025).